

About

How to set up V2Ray

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|---------------------------------------------------------------------------------------------------|------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------|
|  <b>phaitc</b> | docs(app): Updated with 2025 information | 1b363f5 · 8 months ago |  15 Commits |
|  .gitignore    | Initial commit                           |                        | 3 years ago                                                                                      |
|  README.md     | docs(app): Updated with 2025 information |                        | 8 months ago                                                                                     |

# V2ray Server Setup Guide

 **2025 Update:** V2Ray with REALITY protocol remains the most effective method for bypassing the Great Firewall of China. The REALITY protocol eliminates server-side TLS fingerprints and provides superior camouflage compared to traditional VPN protocols, making it significantly harder for the GFW to detect and block.

I tried to install V2Ray on several cloud hosting platforms, including AWS (including Lightsail) and Digital Ocean. However, none of the configurations I experimented with could reliably penetrate China's Great Firewall on AWS servers. Vultr rejected my payment card, so I had to look for alternatives. I chose Digital Ocean. Digital Ocean has been running smoothly for several weeks, with excellent reliability and minimal speed loss.

Using this system has been economical, as each server costs \$4-6 a month (DigitalOcean's entry-level droplets start at \$4/month as of 2025). Although it has the potential for commercial use, my main goal is to provide this information to travelers to China so they can save \$30/month on services like Astrill that often require switching between servers. Both Astrill and V2Ray worked in my experience, but I found the V2Ray setup to be more reliable.

However, there are some drawbacks compared to other VPNs like Astrill, such as the lack of URL-specific bypasses. This may not be a big issue for me while traveling in China, but it may be a problem if you plan to live here for a long time.

If you want to avoid VPNs altogether, the easiest way is to use a SIM card from Hong Kong (if you're flying in from there) or roam with a SIM card from your home country. For example, if you have a SIM card from Malaysia, you can use that SIM card to roam in China while getting 2GB of data per day for only 99RM (\$21 US) per month. This may be cheaper than getting a SIM card in China. However, without a local Chinese phone number, you won't be able to use some local services such as ordering takeout and buying tickets via WeChat.

## Important Notice (2025 Update)

 **Digital Ocean Access from China:** The DigitalOcean website and control panel are blocked in China, making it difficult to create new accounts or manage servers from within China without a VPN. However, **existing droplets set up before entering China can still be accessed and work fine**. It's strongly recommended to set up and test your servers before traveling to China.

## Recommended Server Providers (2025)

1. **Vultr** - Better connectivity to China, multiple Asian locations
2. **Linode** - Good Asian presence, reliable performance
3. **Oracle Cloud** - Offers free tier (limited resources)
4. **BandwagonHost** - Popular among Chinese users
5. **AWS Lightsail** - Works but less reliable for GFW bypass
6. **Google Cloud Platform** - Free tier available, variable success rate

**Note:** Server location matters! Choose servers in:

- Japan (lowest latency for most of China)
- Singapore
- South Korea
- US West Coast (as backup)

## Setting up a V2ray Server

### Digital Ocean Account Setup

1. **Create a Digital Ocean Account:**

- Sign up for a Digital Ocean account using a referral link, like this one: [Digital Ocean Referral Link](#) this will give you \$200 of credit to be used within 2 months.
- Note: Students can get a \$200 credit that is valid for a year via GitHub Education.

2. **Add Billing Information:**

- Add your billing information by linking a credit card to your Digital Ocean account. While other cloud providers can be used, AWS is not recommended due to potential issues.

### Droplet Creation and Configuration

3. **Create a Droplet:**

- In the top right-hand corner, click "Create" and then select "Droplets."

4. **Configure Server Location:**

- Choose a server location that's closest to where you'll be in China. For example, if you're in the eastern area of China, select San Francisco. You can create multiple servers in different locations if needed.

5. **Select OS and Version:**

- Choose "Ubuntu" as the OS image and select "Ubuntu 22.04 LTS (x64)" as the version.

6. **Choose Virtual Machine Size:**

- Opt for "Shared CPU: Basic."

7. **CPU Options:**

- Select "Premium AMD." If you plan to use it with just a few devices (up to 8), choose the entry-level plan starting at \$4/month (512MB RAM) or \$6/month (1GB RAM).

- o Select "Password" for now, or you can set up SSH authentication separately.

### 9. Set Hostname:

- Change the hostname to something of your choice, for organizational purposes.

- Click "Create" and wait for your server to be provisioned.

- o After the server is created, note down the IP address as you'll need it later.

## 12. Access the Server Console:

- frankfurt-1** in  **V2Ray** / 1 GB Memory / 1 AMD vCPU / 25 GB Disk / FRA1 - Ubuntu 22.04 (LTS) x64

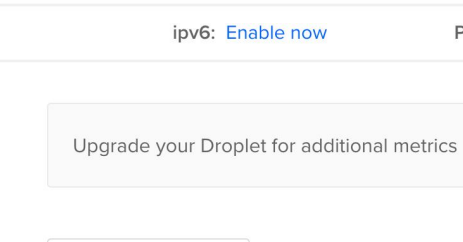
ipv4:  ipv6: [Enable now](#) Private IP: 10.114.0.2 Reserved IP: [Enable now](#) [Console: !\[\]\(5890ff4c38007932c846fa9d39ba1fe6\_img.jpg\)](#)

[Graphs](#)  
[Access](#)  
[Power](#)  
[Volumes](#)  
[Resize](#)  
[Networking](#)  
[Backups](#)  
[Snapshots](#)  
[Kernel](#)  
[History](#)  
[Destroy](#)  
[Tags](#)  
[Recovery](#)

Upgrade your Droplet for additional metrics and alerting.

Select period  
1 hour 

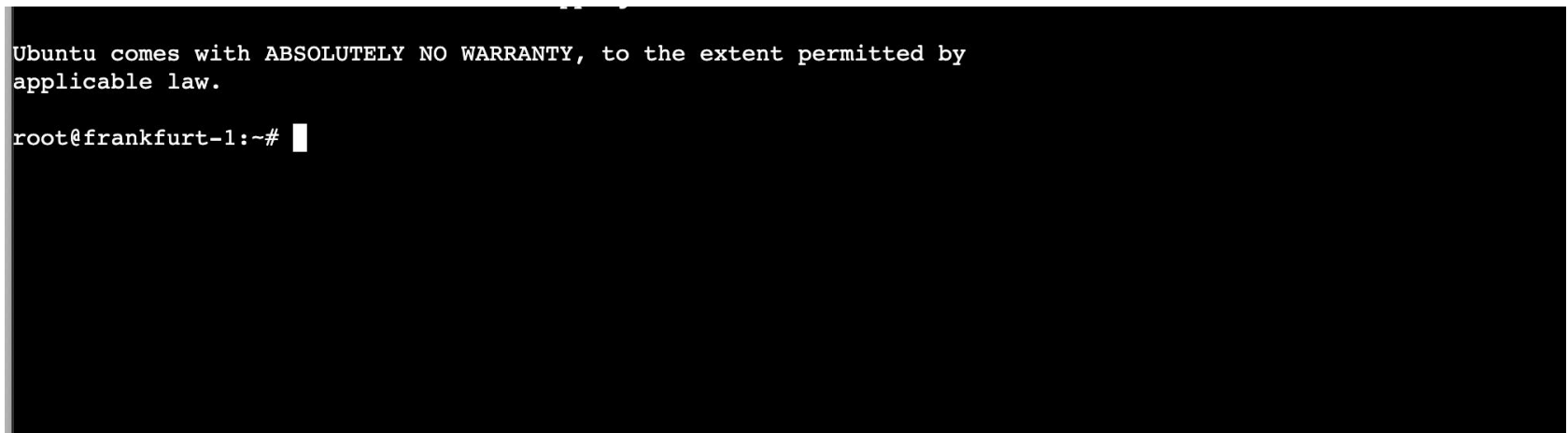
**Bandwidth**



The graph displays bandwidth usage over a 1-hour period. The y-axis ranges from -1.0 b/s to 1.0 b/s. The x-axis shows time from 12:10 to 13:05. The graph area is empty with the text "No data".

This Droplet now has our updated Droplet console with better performance and more features

OK



- o Run the command `export DEBIAN_FRONTEND=noninteractive && apt update -y && apt upgrade -y` to update the server with the latest packages.

- o For better security and active maintenance, use 3x-ui: `bash <(curl -Ls https://raw.githubusercontent.com/mhsanaei/3x-ui/master/install.sh)`
- o Alternative (original x-ui): `bash <(curl -Ls https://raw.githubusercontent.com/FranzKafkaYu/x-ui/master/install_en.sh)`

- Follow the prompts for X-Ray installation:
  - Type "y" to continue.

16. Enable BBR:

- Type "x-ui" and press Enter.
- Type "15" and press Enter to enable BBR.

Web-Based Admin Panel Access

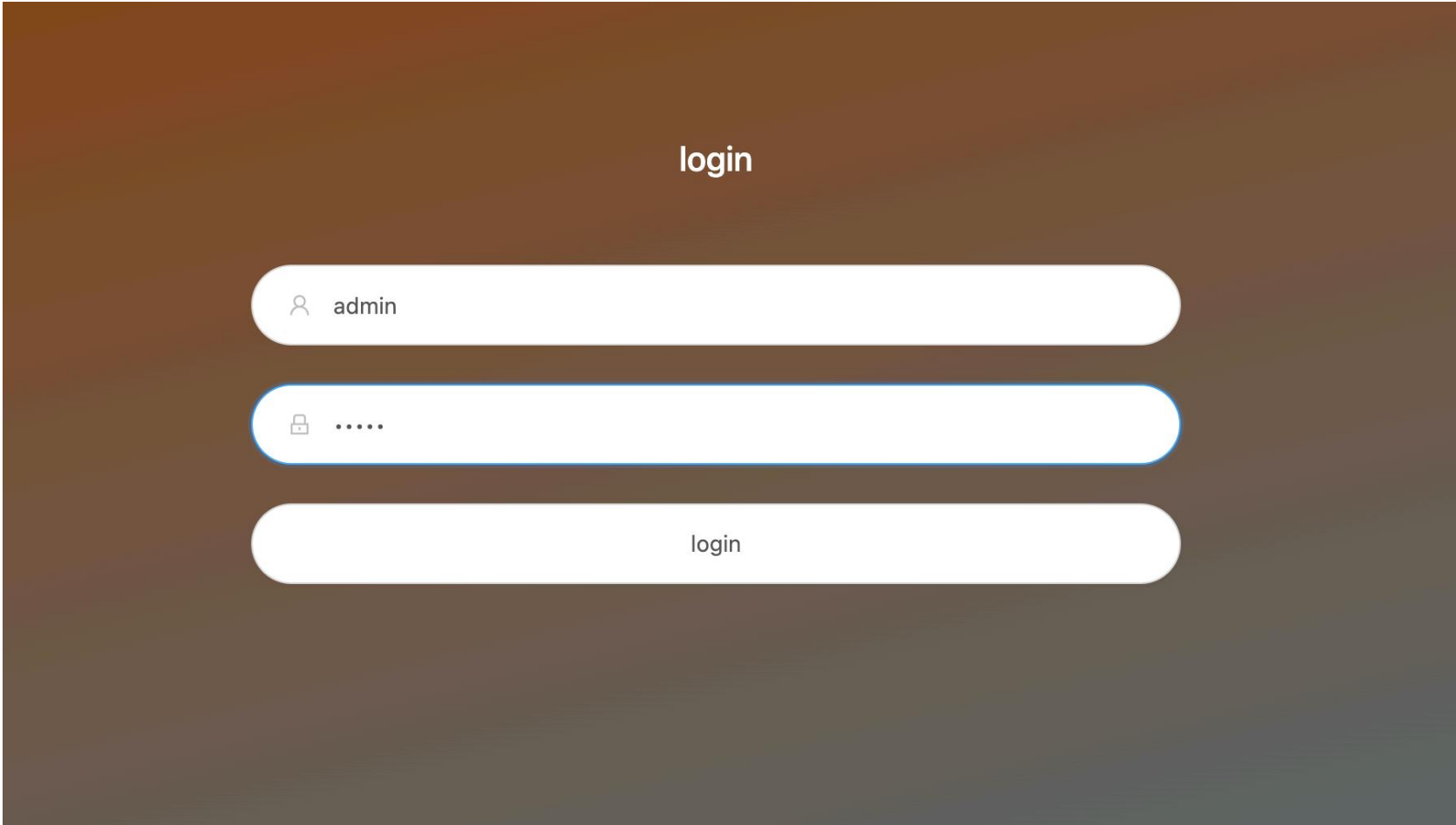
17. Access Admin Panel in a Web Browser:

- Open a web browser and enter your server's IP address followed by the port number (e.g., "http://your\_ip\_address:54321").



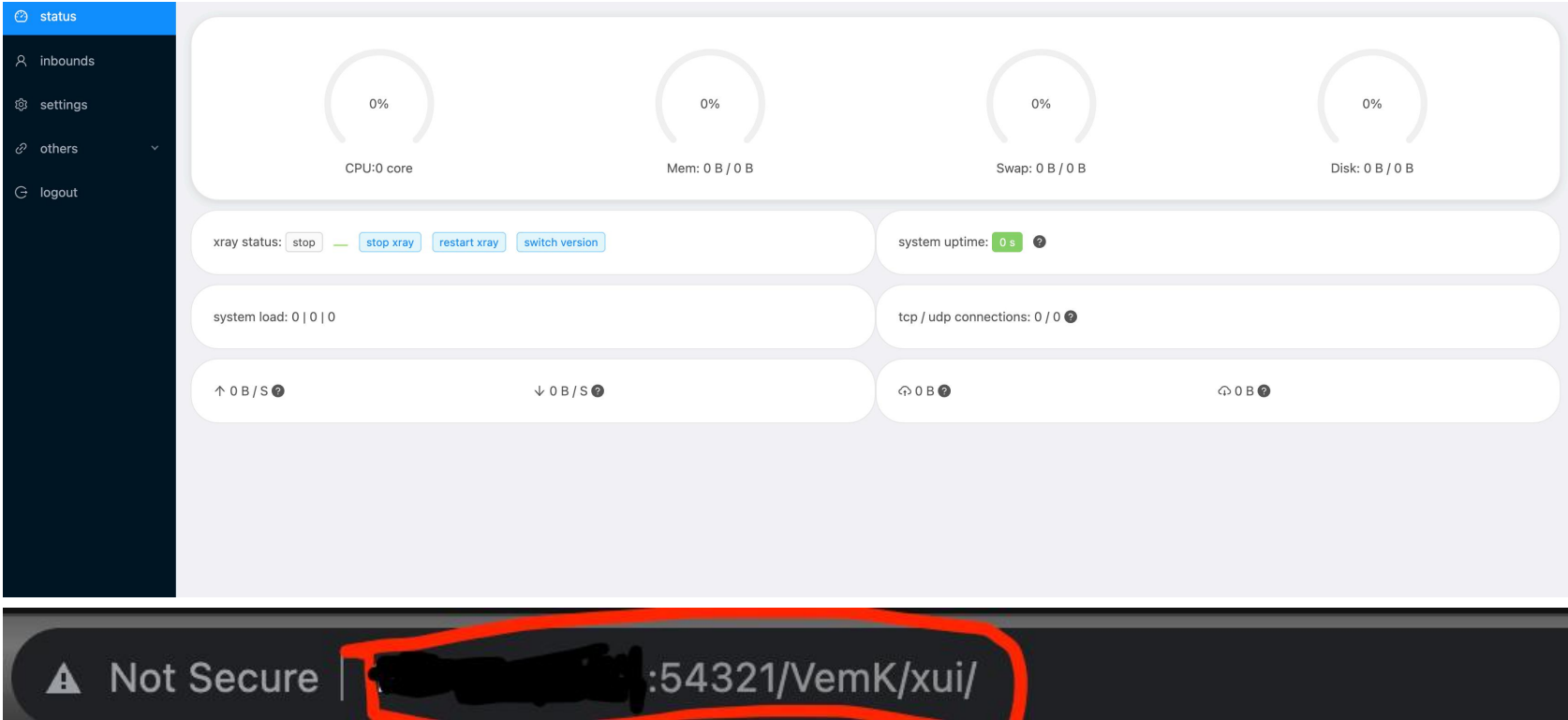
18. Log in to X-UI Admin Panel:

- Use the username and password you set up in step 15 to log in.



19. Admin Panel Settings:

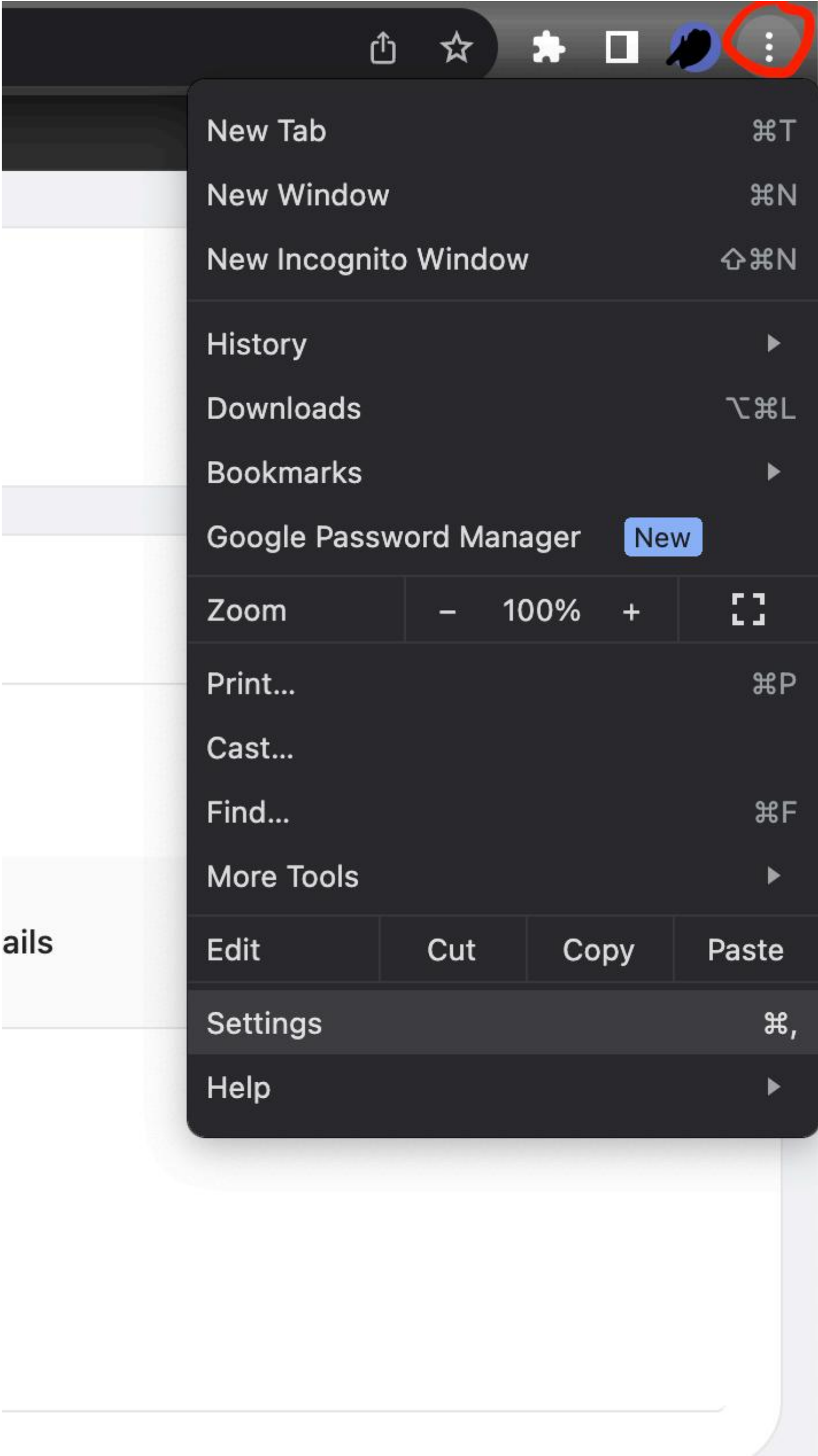
- Click on "Settings" and confirm the panel restart. Note the new URL for accessing the admin panel.

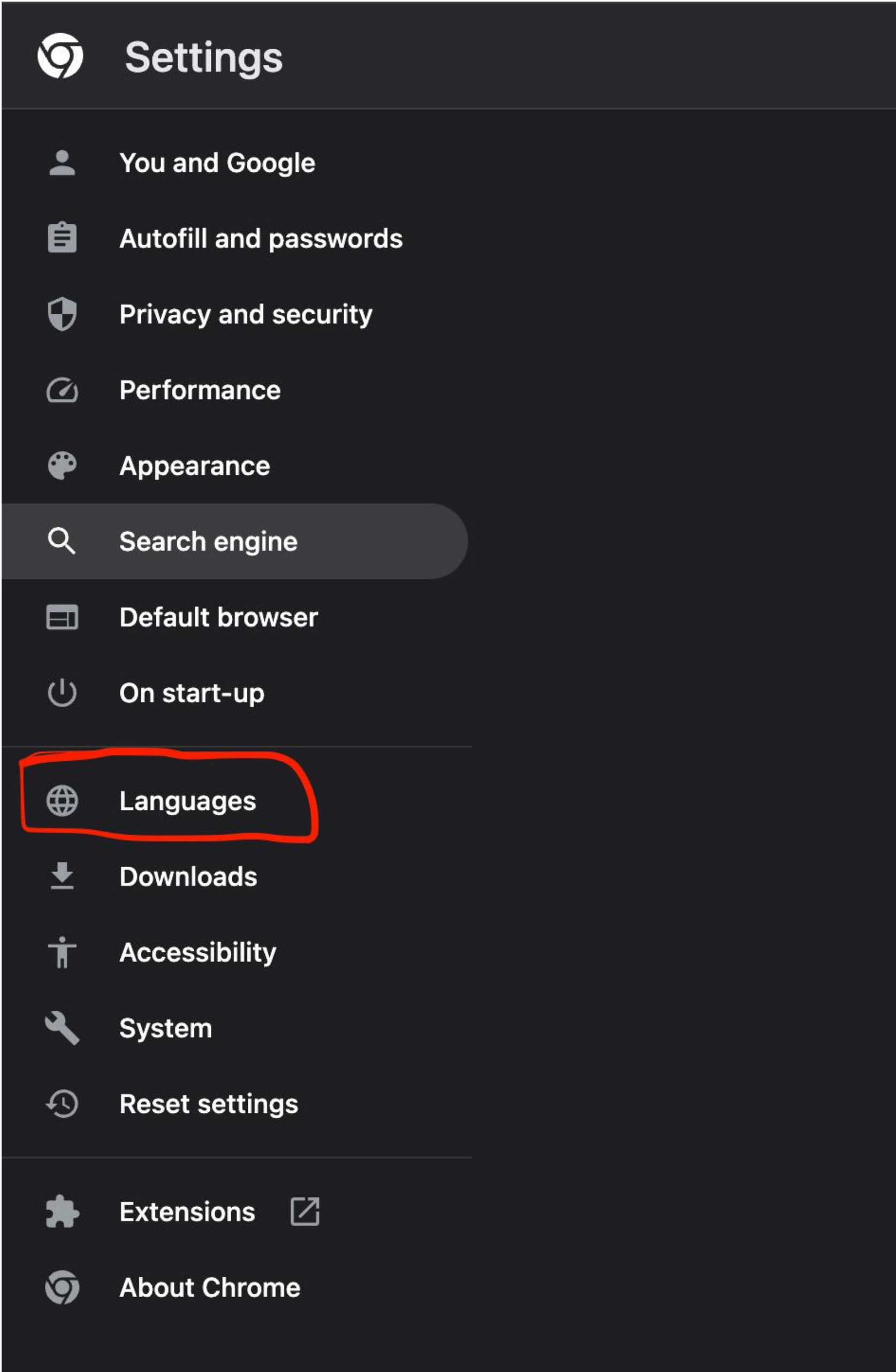


Inbound Rules Configuration

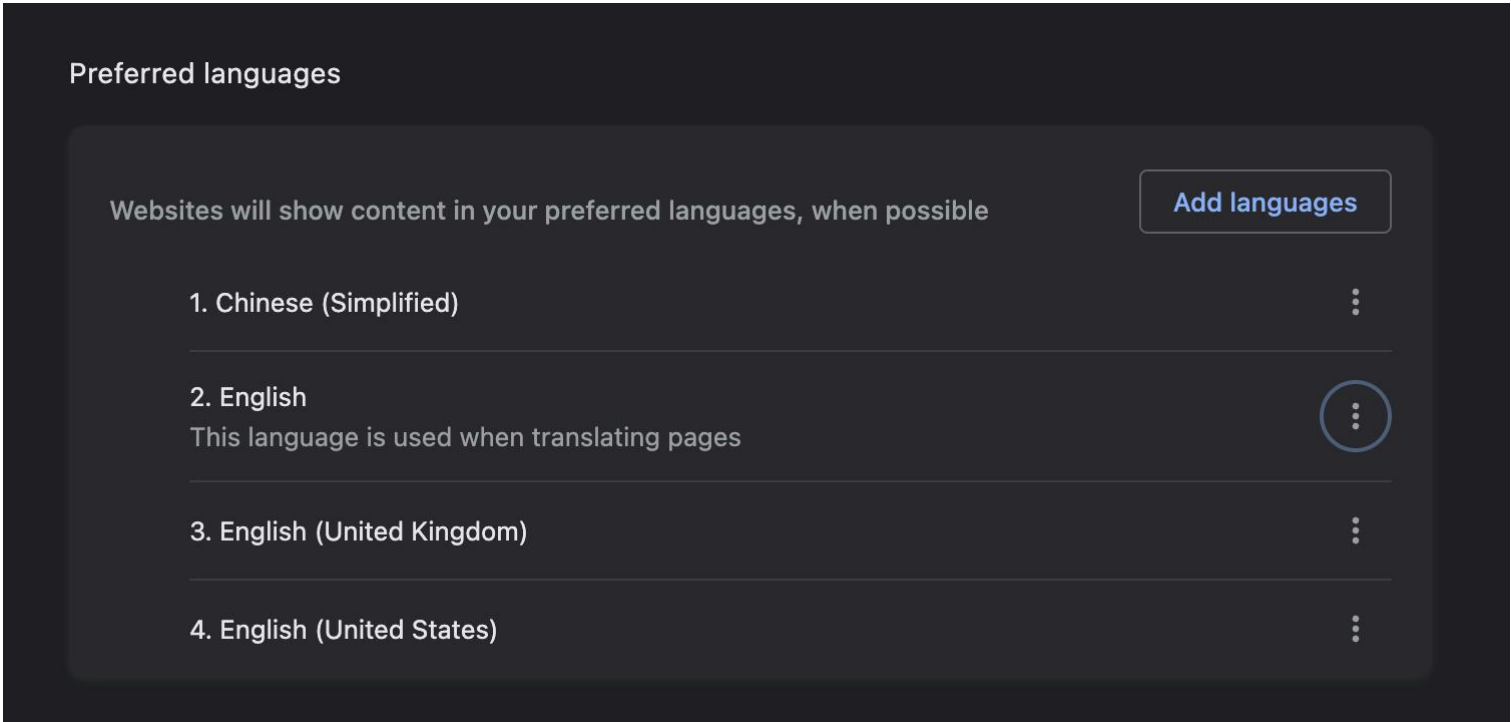
20. Preferred Language Setup:

- In your web browser settings, set "Chinese (Simplified)" as the preferred language for adding an inbound. Panel is a little buggy and you won't be able to add inbounds without doing this first.









21. Add Inbound Rules (REALITY Protocol - Recommended for 2025):

- Refresh the admin panel and go to "Inbound" -> "Add inbound."
- Use the settings for "VLESS + XTLS + uTLS + REALITY" (most effective for bypassing GFW in 2025)
- Set the remark to your hostname (Step 8) and port to "443."
- Toggle to enable "reality."
- Choose a camouflage website (e.g., [www.amazon.com](https://www.amazon.com), dl.google.com) that supports TLSv1.3

*Note: Don't worry if your information below the "reality" row is different to the screenshot as those are automatically populated.*

Add inbound

X

remark: frankfurt-1

enable: ☒

protocol: vless

listening ip ?:

port: 443

total ?:

0

expiryTime ?:

Select date

auto reset ?:

device alert ?:

device limit ?:

0

add user:

+

+3

+5

+10

fallbacks:

+

network: tcp

acceptProxyProtocol:

http camouflage:

reality ?:

tls:

xtls ?:

show:

xver:

0

dest: www.speedtest.org:443

serverNames: www.speedtest.org

privateKey:

publicKey:

shortIds: ,41,18de,172847,19c486dc

sniffing ?:

☒ http

☒ tls

☒ quic



22. Add a User:

- Click the "+" button next to "Add user" and select "xtls-rprx-vision" as the flow.

add user: 

+

+3

+5

+10

User1

id: 27bdec13-97b1-4d

flow: xtls-rprx-vision

utls: chrome

email: g3BV.love@xray.cc

expiry ? : Select date

traffic ? : 0

usage: 

↑NaN PB | NaN PB↓

used : NaN PB

23. Create the Inbound:

- Click "Add" at the bottom of the page to create your first inbound.

up / down: 0 B / 0 B

total: 0 B

inbounds: 1

Add Inbound

Reset Traffic

Export Links

search

| Operation | Enable      | Id | Remark      | Protocol | Port | Traffic↑ ↓                                | Details          | Transport                         | ExpiryTime           |
|-----------|-------------|----|-------------|----------|------|-------------------------------------------|------------------|-----------------------------------|----------------------|
| Operation | <div></div> | 1  | frankfurt-1 | vless    | 443  | <div>0 B / 0 B</div> <div>unlimited</div> | <div>check</div> | <div>tcp</div> <div>reality</div> | <div>unlimited</div> |

24. Update X-Ray:

- In the X-UI admin panel, switch to the newest version of X-Ray on the status page and click "Restart X-ray."

status

inbounds

settings

others

logout

0.5%

CPU: 1 core

18.98%

Mem: 184.02 MB / 969.45 MB

0%

Swap: 0 B / 0 B

8.4%

Disk: 2.02 GB / 24.05 GB

xray status: 

running

1.7.5

stop xray

restart xray

switch version

system uptime: 53 minutes

system load: 0 | 0 | 0

tcp / udp connections: 13 / 3

↑ 0 B / S

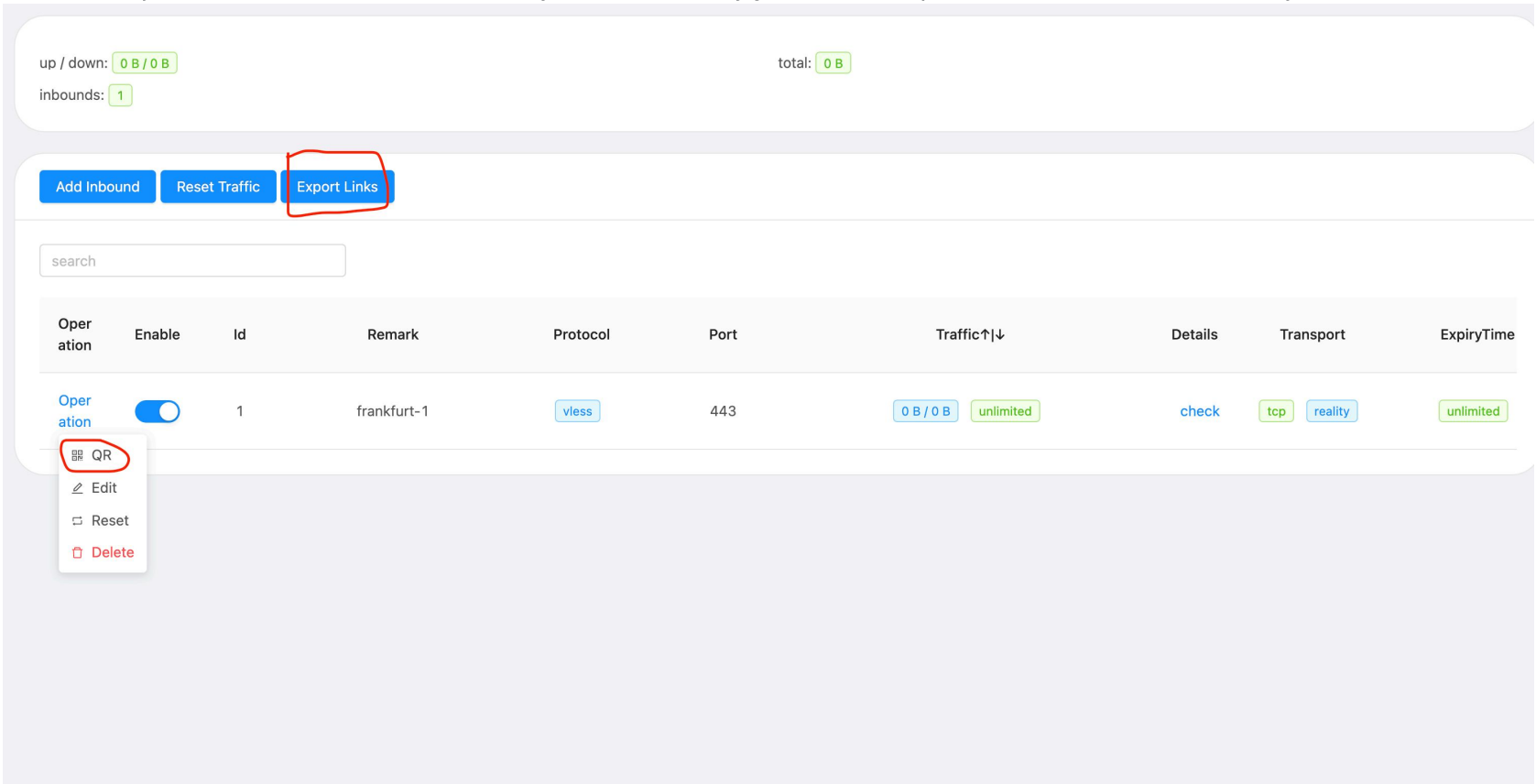
↓ 0 B / S

6.73 MB

175.53 MB

25. Export Links:

- o You can export links and share them with your users. They just need to paste the link into their V2ray client.



## Alternative Circumvention Methods (2025 Update)

As of 2025, the Great Firewall has become more sophisticated. Here are alternative and complementary methods:

### 1. Protocol Alternatives

- Shadowsocks with obfuscation plugins - Still effective with proper configuration
- Trojan-GFW - Mimics HTTPS traffic effectively
- WireGuard - Fast and lightweight, though requires careful configuration
- Hysteria/Hysteria2 - UDP-based protocol optimized for unstable connections

### 2. Free Tools (Use with Caution)

- Freegate - Developed by Falun Gong practitioners, still actively maintained
- Psiphon - Open-source circumvention tool
- Lantern - P2P censorship circumvention tool
- Tor with Snowflake bridges - Requires specific configuration for China